R22

Code No.: 22CSC35

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. & B.Tech III Sem (Main) Examination Dec 2023 – Jan 2024

Data Structures using Python

(Common to Mech, Chem & Biotech)

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A
(5Q X 2M = 10 Marks)

		M	CO	BT
1	What is Data Visualization?	(2)	2	1
2	Define sorting and write the time complexities of selection sort, merge sort, quick sort and radix sort.	(2)	3	2
3	Differentiate between singly linked list and doubly linked list	(2)	4	3
4	List out applications of graph.	(2)	4	3
5	Define Hash table?	(2)	4	1

Part - B (50 Marks)
(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Explain Class and Object in python with an example.	(5)	1	1
	(b)	What is NumPy? Create 2D array and 3D arrays using NumPy.	(5)	2	3
		(OR)			
7	(a)	Write briefly about pandas series and Write program to create pandas series for the given NumPy array A=[20,47,12, 63, 58, 75]	(5)	2	3
	(b)	Explain selection and slicing in pandas with examples.	(5)	2	2
8	(a)	What is Data structure? Discuss briefly about classification of data structures.	(5)	1	1
	(b)	Explain Asymptotic Notations with neat sketch.	(5)	1	2
		(OR)			
9	(a)	Using radix sort technique, sort the following list. 4245,114,2745,12, 6578,34,9876,143,341,7777	(5)	3	3
	(b)	Write python program to Implement Quick sort.	(5)	3	1
10	(a)	Write an algorithm/pseudocode to count the number of nodes in a singly linked list.	(5)	4	2
	(b)	Illustrate the procedure to insert a node at the beginning of doubly linked list.	(5)	4	2
		(OR)			
11	(a)	Illustrate Push and Pop operations of stack using linked list.	(5)	4	2
	(b)	Which data structure you would choose out of STACK and QUEUE for railway ticket reservation? Justify your choice.	(5)	4	4

12	(a)	Illustrate the procedure to delete node with two children (left child and right child) from binary search tree with an example.	(5)	4	3
	(b)	Discuss the Properties of Binary Trees.	(5)	4	1
		(OR)			
13	(a)	Illustrate the representation of graph using Adjacency matrix.	(5)	4	2
	(b)	Give algorithm for DFS and demonstrate DFS using suitable example.	(5)	4	3
14	(a)	Describe the brute force algorithm for pattern matching.	(5)	5	1
	(b)	Solve the Rabin-Karp algorithm for the following Example: Text: HEREISANEXAMPLE Pattern: EXAMPLE	(5)	5	3
		(OR)			
15	(a)	Resolve collisions by using linear probing technique for hash table of size 11 for the following input 22, 15, 78, 77, 51, 32, 65, 11.	(5)	4	3
	(b)	Discuss briefly about separate chaining technique.	(5)	4	1

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